

Document Intelligence and Retrieval-Augmented Generation

An 8-page two-column test corpus for LightningParse

This synthetic technical paper is designed as a controlled PDF test corpus. It contains headings, subheadings, dense paragraphs, repeated terminology, a benchmark table, and references to page and section structure. The goal is to create a document that resembles the kind of technical material commonly placed into a retrieval-augmented generation pipeline.

1. The Shape of a Modern Computing System

A modern computing system is rarely a single program performing one isolated task. It is better understood as a layered collection of components that cooperate across different time scales. A user request may pass through an interface, an application service, a cache, a database, a message queue, and several specialized workers before a result is returned.

Layers and boundaries

The value of a layered design is not merely organizational. Boundaries make assumptions visible. A storage layer can expose durability guarantees without exposing the physical layout of its files. An API can define an input contract without revealing the implementation behind it. A parser can expose structured document blocks while hiding the details of character decoding, layout reconstruction, and page traversal.

Why boundaries matter

Good boundaries also make experiments easier. If a document parser is isolated behind a small interface, the parser can be replaced without rewriting the retrieval system around it. This property is especially valuable in information retrieval systems, where the quality of upstream document processing can influence every downstream stage.

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2. Documents Are More Than Text

A PDF page may look like a simple sequence of paragraphs to a human reader, but internally it can contain positioned glyphs, drawing instructions, images, annotations, tables, headers, footers, and multiple columns. Reading order therefore becomes an inference problem. A naive text extractor may return characters correctly while still producing a sequence that is confusing or misleading.

Reading order

Consider a two-column research article. A human reader naturally finishes the left column before moving to the right column. A simplistic extractor might interleave lines from both columns, producing sentences that never existed in the original document. The problem is not character recognition; it is reconstruction of document structure.

Metadata as context

Page numbers, section identifiers, captions, and source filenames are useful context. Preserving this information allows later systems to explain where a retrieved passage came from. For a retrieval-augmented generation system, this can turn an opaque answer into an auditable answer with a clear document trail.

Layout reconstruction

Layout-aware extraction becomes particularly important when a document contains sidebars, footnotes, equations, captions, or multiple reading regions. A robust parser must distinguish visual proximity from logical sequence. Two blocks can be physically close while belonging to completely different semantic regions.

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3. Retrieval-Augmented Generation

Retrieval-augmented generation, commonly abbreviated RAG, combines information retrieval with language generation. Instead of asking a language model to answer solely from its learned parameters, a system first retrieves relevant passages from a document collection. Those passages are then provided as context to the model.

Indexing

The indexing stage usually begins by splitting documents into chunks. Each chunk is converted into a numerical embedding, and the resulting vectors are stored in a vector index. A query is embedded using the same representation, and the index returns chunks whose vectors are close to the query vector.

Why chunking matters

Chunk size creates a trade-off. Very small chunks can lose context and produce fragmented evidence. Very large chunks can dilute relevance and consume more context-window capacity. Structure-aware chunking can be more useful than blindly splitting every document after a fixed number of characters.

Retrieval context

A retrieved passage is useful only when it contains enough surrounding information to answer the question. Section headings, page identifiers, captions, and document names can therefore become part of the retrieval metadata even when they are not embedded directly.

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4. Semantic Search and Embeddings

An embedding model maps text into a high-dimensional numerical space. Texts with related meanings tend to occupy nearby regions of that space. This allows a retrieval system to find passages that are semantically related even when they do not share many exact keywords.

Similarity

A common retrieval strategy uses cosine similarity. If two normalized vectors point in similar directions, their cosine similarity is high. In practice, an index may use optimized approximate-nearest-neighbor techniques so that searches remain fast as the number of stored chunks grows.

Failure cases

Semantic similarity is not the same thing as factual correctness. A retrieved passage can be topically related but fail to contain the answer. Retrieval evaluation should therefore consider whether the required evidence was actually returned, not merely whether the retrieved text sounded relevant.

Hybrid retrieval

Keyword retrieval and semantic retrieval can complement each other. Exact terms are valuable for identifiers, numbers, names, and technical expressions, while embeddings are useful when the wording of a query differs from the wording of the source. A production system may combine both signals.

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5. Tables, Figures, and Structured Evidence

Tables are particularly important because they often contain the most information-dense parts of technical and scientific documents. A table may visually communicate relationships through rows and columns that disappear when the page is converted into a flat text stream.

Structured representation

A useful document-processing pipeline can preserve a table as rows and columns before converting it into a representation suitable for retrieval. Markdown is one practical intermediate format because it preserves the basic relationship between headers and cells while remaining easy for language models to interpret.

Benchmark example

Suppose a benchmark reports four systems and their accuracy, precision, and recall. A retrieval system should allow a user to ask which system achieved the highest recall and recover both the system name and the associated value. Flattening the table incorrectly can make such questions surprisingly difficult.

Table for extraction testing

The following synthetic table is deliberately included to test whether a parser can preserve headers, rows, numeric values, and their relationships.

System	Accuracy	Precision	Recall
Atlas	91.4%	89.8%	88.7%
Borealis	94.2%	93.5%	92.1%
Cirrus	96.1%	95.0%	95.4%
Draco	93.7%	94.1%	90.8%

Table 1. Synthetic benchmark values for table-extraction testing.

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6. Concurrency and Performance

Document ingestion can become expensive when a system processes hundreds or thousands of files. Parsing, OCR, image processing, tokenization, and embedding generation may each consume significant CPU or memory resources. The total ingestion time is therefore determined by the slowest stages and by how effectively independent work can be performed concurrently.

Native code

A Python application can delegate compute-intensive work to a native implementation when appropriate. Rust is attractive for this role because it provides strong memory-safety guarantees and predictable performance while still supporting interoperability with Python through foreign-function interfaces.

Measuring the right thing

A useful benchmark should separate cold-start effects from steady-state performance. It should record document count, page count, parsing time, throughput, memory consumption, and error rates. A single headline number can be impressive while hiding a workload that does not represent realistic usage.

Concurrency experiment

For a parser benchmark, independent documents are natural units of parallel work. The experiment should report both sequential and concurrent execution, while also checking that outputs remain identical. Speed without correctness is not a useful optimization.

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7. Evaluating a Document Pipeline

A parser that is extremely fast but systematically damages reading order may reduce the quality of a downstream RAG system. Evaluation should therefore include both engineering metrics and information-quality metrics.

Suggested measurements

Useful measurements include extraction latency per page, number of recovered blocks, chunk count, average chunk length, retrieval latency, retrieval recall at a chosen k , and answer accuracy on a small set of questions whose answers are known.

Controlled comparison

A fair experiment keeps the corpus, embedding model, chunking policy, retrieval configuration, and questions constant while changing only the parser. This makes it easier to attribute observed differences to the document-processing stage rather than to unrelated changes elsewhere.

Error analysis

When a question receives an incorrect answer, inspect the complete chain: extraction, structure detection, chunking, embedding, retrieval, and generation. A useful benchmark should make it possible to identify the first stage at which the required evidence was lost.

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8. From Prototype to Reliable System

A reliable prototype does not need a large infrastructure stack. A local vector index, a small embedding model, and a handful of representative documents are enough to reveal many problems. The goal of the first experiment is to discover where the pipeline fails, not to optimize a system that has not yet been understood.

Traceability

Every retrieved chunk should retain its source document and location. This makes debugging dramatically easier. When an answer is wrong, the developer can inspect the exact retrieved evidence and determine whether the failure came from parsing, chunking, retrieval, or generation.

The larger lesson

RAG quality is an end-to-end property. Better generation cannot fully compensate for corrupted extraction, poor chunk boundaries, or missing evidence. Conversely, a strong document representation gives every downstream stage better material to work with. The most useful systems therefore treat document processing as a first-class engineering problem rather than a preprocessing afterthought.

Final experiment

The recommended test is simple: parse this document with LightningParse, inspect its page and block structure, create metadata-aware chunks, embed them, retrieve relevant passages, and ask questions about both prose and the table. Then repeat the same experiment with a baseline PDF extractor and compare the results.

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